

- 7 The region R of the Argand diagram consists of the set of points representing complex numbers z which satisfy the following inequalities.

$$\operatorname{Im}(z) \geq 0 \qquad \arg(z+2) \leq \frac{1}{4}\pi \qquad |z| \leq |z-4-2i|$$

- (a) Sketch and clearly label the region R on an Argand diagram. [4]
- (b) **In this question you must show detailed reasoning.**

Find the largest value of $\operatorname{Im}(z)$ in the region R. [7]